

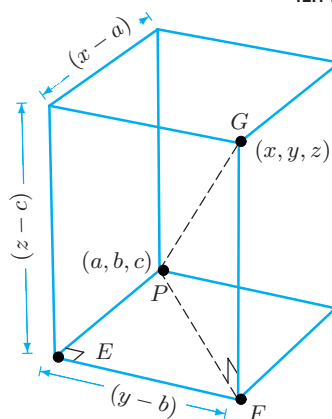


Figure 12.10: The diagonal PG gives the distance between the points (x, y, z) and (a, b, c)

Using Pythagoras' theorem twice gives

$$(PG)^2 = (PF)^2 + (FG)^2 = (PE)^2 + (EF)^2 + (FG)^2 = (x-a)^2 + (y-b)^2 + (z-c)^2.$$

Thus, a formula for the distance between the points (x, y, z) and (a, b, c) in 3-space is

$$\text{Distance} = \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2}.$$

Example 11 Find the distance between $(1, 2, 1)$ and $(-3, 1, 2)$.

Solution Distance $= \sqrt{(-3-1)^2 + (1-2)^2 + (2-1)^2} = \sqrt{18} = 4.24$.

Example 12 Find an expression for the distance from the origin to the point (x, y, z) .

Solution The origin has coordinates $(0, 0, 0)$, so the distance from the origin to (x, y, z) is given by

$$\text{Distance} = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2} = \sqrt{x^2 + y^2 + z^2}.$$

Example 13 Find an equation for a sphere of radius 1 with center at the origin.

Solution The sphere consists of all points (x, y, z) whose distance from the origin is 1, that is, which satisfy the equation

$$\sqrt{x^2 + y^2 + z^2} = 1.$$

This is an equation for the sphere. If we square both sides we get the equation in the form

$$x^2 + y^2 + z^2 = 1.$$

Note that this equation represents the *surface* of the sphere. The solid ball enclosed by the sphere is represented by the inequality $x^2 + y^2 + z^2 \leq 1$.

Exercises and Problems for Section 12.1

Exercises

- Which of the points $P = (1, 2, 1)$ and $Q = (2, 0, 0)$ is closest to the origin?
- Which two of the three points $P_1 = (1, 2, 3)$, $P_2 = (3, 2, 1)$ and $P_3 = (1, 1, 0)$ are closest to each other?

3. Which of the points $A = (1.3, -2.7, 0)$, $B = (0.9, 0, 3.2)$, $C = (2.5, 0.1, -0.3)$ is closest to the yz -plane? Which one lies on the xz -plane? Which one is farthest from the xy -plane?
4. You are at the point $(3, 1, 1)$, standing upright and facing the yz -plane. You walk 2 units forward, turn left, and walk another 2 units. What is your final position? From the point of view of an observer looking at the coordinate system in Figure 12.2 on page 668, are you in front of or behind the yz -plane? To the left or to the right of the xz -plane? Above or below the xy -plane?
5. On a set of x , y and z axes oriented as in Figure 12.5 on page 669, draw a straight line through the origin, lying in the yz -plane and such that if you move along the line with your y -coordinate increasing, your z -coordinate is increasing.
6. What is the midpoint of the line segment joining the points $(-1, 3, 9)$ and $(5, 6, -3)$?
7. Find the equation of the sphere with radius 2 and centered at $(1, 0, 0)$.
8. Find the equation of the vertical plane perpendicular to the y -axis and through the point $(2, 3, 4)$.

Exercises 15–17 refer to the map in Figure 12.1 on page 666.

In Exercises 7–10 sketch graphs of the equations in 3-space.

7. $z = 4$
8. $x = -3$
9. $y = 1$
10. $z = 2$ and $y = 4$
11. With the z -axis vertical, a sphere has center $(2, 3, 7)$ and lowest point $(2, 3, -1)$. What is the highest point on the sphere?
12. Find an equation of the sphere with radius 5 centered at the origin.
15. Give the range of daily high temperatures for:
 - (a) Pennsylvania
 - (b) North Dakota
 - (c) California
16. Sketch a possible graph of the predicted high temperature T on a line north-south through Topeka.
17. Sketch possible graphs of the predicted high temperature on a north-south line and an east-west line through Boise.

For Exercises 18–20, refer to Table 12.1 on page 667, where p is the price of beef and I is annual household income.

18. Give tables for beef consumption as a function of p , with I fixed at $I = 20$ and $I = 100$. Give tables for beef consumption as a function of I , with p fixed at $p = 3.00$ and $p = 4.00$. Comment on what you see in the tables.
19. Make a table of the proportion, P , of household income spent on beef per week as a function of price and income. (Note that P is the fraction of income spent on beef.)
20. How does beef consumption vary as a function of household income if the price of beef is held constant?

Problems

21. The temperature adjusted for wind chill is a temperature which tells you how cold it feels, as a result of the combination of wind and temperature.² See Table 12.2.

Table 12.2 Temperature adjusted for wind chill ($^{\circ}\text{F}$) as a function of wind speed and temperature

		Temperature ($^{\circ}\text{F}$)							
		35	30	25	20	15	10	5	0
Wind Speed (mph)	5	31	25	19	13	7	1	-5	-11
	10	27	21	15	9	3	-4	-10	-16
	15	25	19	13	6	0	-7	-13	-19
	20	24	17	11	4	-2	-9	-15	-22
	25	23	16	9	3	-4	-11	-17	-24

- (a) If the temperature is 0°F and the wind speed is 15 mph, how cold does it feel?
- (b) If the temperature is 35°F , what wind speed makes it feel like 24°F ?
- (c) If the temperature is 25°F , what wind speed makes it feel like 12°F ?

- (d) If the wind is blowing at 20 mph, what temperature feels like 0°F ?

In Problems 22–23, use Table 12.2 to make tables with the given properties.

22. The temperature adjusted for wind chill as a function of wind speed for temperatures of 20°F and 0°F .
23. The temperature adjusted for wind chill as a function of temperature for wind speeds of 5 mph and 20 mph.
24. A car rental company charges \$40 a day and 15 cents a mile for its cars.
 - (a) Write a formula for the cost, C , of renting a car as a function, f , of the number of days, d , and the number of miles driven, m .
 - (b) If $C = f(d, m)$, find $f(5, 300)$ and interpret it.
25. The gravitational force, F newtons, exerted on an object by the earth depends on the object's mass, m kilograms, and its distance, r meters, from the center of the earth, so $F = f(m, r)$. Interpret the following statement in terms of gravitation: $f(100, 7000000) \approx 820$.

²Data from www.nws.noaa.gov/om/windchill, accessed on May 30, 2004.

26. Consider the acceleration due to gravity, g , at a distance h from the center of a planet of mass m .
- If m is held constant, is g an increasing or decreasing function of h ? Why?
 - If h is held constant, is g an increasing or decreasing function of m ? Why?
27. A cube is located such that its top four corners have the coordinates $(-1, -2, 2)$, $(-1, 3, 2)$, $(4, -2, 2)$ and $(4, 3, 2)$. Give the coordinates of the center of the cube.
28. Describe the set of points whose distance from the x -axis is 2.
29. Describe the set of points whose distance from the x -axis equals the distance from the yz -plane.
30. Find a formula for the shortest distance between a point (a, b, c) and the y -axis.
31. Find the equations of planes that just touch the sphere $(x-2)^2 + (y-3)^2 + (z-3)^2 = 16$ and are parallel to
- The xy -plane
 - The yz -plane
 - The xz -plane
32. Find an equation of the largest sphere contained in the cube determined by the planes $x = 2$, $x = 6$; $y = 5$, $y = 9$; and $z = -1$, $z = 3$.
33. A cube has edges parallel to the axes. One corner is at $A = (5, 1, 2)$ and the corner at the other end of the longest diagonal through A is $B = (12, 7, 4)$.
- What are the coordinates of the other three vertices on the bottom face?
 - What are the coordinates of the other three vertices on the top face?
34. Which of the points $P_1 = (-3, 2, 15)$, $P_2 = (0, -10, 0)$, $P_3 = (-6, 5, 3)$ and $P_4 = (-4, 2, 7)$ is closest to $P = (6, 0, 4)$?
35. (a) Find the equations of the circles (if any) where the sphere $(x-1)^2 + (y+3)^2 + (z-2)^2 = 4$ intersects each coordinate plane.
(b) Find the points (if any) where this sphere intersects each coordinate axis.
36. A rectangular solid lies with its length parallel to the y -axis, and its top and bottom faces parallel to the plane $z = 0$. If the center of the object is at $(1, 1, -2)$ and it has a length of 13, a height of 5 and a width of 6, give the coordinates of all eight corners and draw the figure labeling the eight corners.
37. An equilateral triangle is standing vertically with a vertex above the xy -plane and its two other vertices at $(7, 0, 0)$ and $(9, 0, 0)$. What is its highest point?
38. (a) Find the midpoint of the line segment joining $A = (1, 5, 7)$ to $B = (5, 13, 19)$.
(b) Find the point one quarter of the way along the line segment from A to B .
(c) Find the point one quarter of the way along the line segment from B to A .

Strengthen Your Understanding

In Problems 39–41, explain what is wrong with the statement.

39. In 3-space, $y = 1$ is a line parallel to the x -axis.
 40. The xy -plane has equation $xy = 0$.
 41. The distance from $(2, 3, 4)$ to the x -axis is 2.

In Problems 42–43, give an example of:

42. A formula for a function $f(x, y)$ that is increasing in x and decreasing in y .
 43. A point in 3-space with all its coordinates negative and farther from the xz -plane than from the plane $z = -5$.

Are the statements in Problems 44–57 true or false? Give reasons for your answer.

44. If $f(x, y)$ is a function of two variables defined for all x and y , then $f(10, y)$ is a function of one variable.
 45. The volume V of a box of height h and square base of side length s is a function of h and s .
 46. If $H = f(t, d)$ is the function giving the water temperature $H^\circ\text{C}$ of a lake at time t hours after midnight and depth d meters, then t is a function of d and H .
 47. A table for a function $f(x, y)$ cannot have any values of f appearing twice.
48. If $f(x)$ and $g(y)$ are both functions of a single variable, then the product $f(x) \cdot g(y)$ is a function of two variables.
 49. The point $(1, 2, 3)$ lies above the plane $z = 2$.
 50. The graph of the equation $z = 2$ is a plane parallel to the xz -plane.
 51. The points $(1, 0, 1)$ and $(0, -1, 1)$ are the same distance from the origin.
 52. The point $(2, -1, 3)$ lies on the graph of the sphere $(x-2)^2 + (y+1)^2 + (z-3)^2 = 25$.
 53. There is only one point in the yz -plane that is a distance 3 from the point $(3, 0, 0)$.
 54. There is only one point in the yz -plane that is distance 5 from the point $(3, 0, 0)$.
 55. If the point $(0, b, 0)$ has distance 4 from the plane $y = 0$, then b must be 4.
 56. A line parallel to the z -axis can intersect the graph of $f(x, y)$ at most once.
 57. A line parallel to the y -axis can intersect the graph of $f(x, y)$ at most once.